

SOC SECURITY INCIDENT REPORT: DNS ANOMALY & THREAT HUNTING

Incident Reference: INC-2026-0822-DNS

Date: August 22, 2026

Investigator: Tier 1 SOC Analyst / Threat Hunter

Status: Closed / Remediated & Tuned

Severity: Medium / High

1. Executive Summary

During proactive threat hunting operations, security analysts ingested and scrutinized **422,130 Zeek/Bro DNS transaction events** within Splunk Enterprise. The operation identified internal reconnaissance activity, high-volume beaconing patterns, and potential exfiltration anomalies. Root-cause analysis successfully differentiated between malicious behavior, benign protocol noise, and local environment false positives.

2. Incident Scope & Impacted Assets

IP Address	Host Role	Activity Identified	Verdict
10.10.117.210	Internal Host	Top DNS consumer (75,943 queries); heavy Teredo tunneling traffic (27,425 queries).	Benign / Protocol Overhead (Monitoring Required)
192.168.202.83	Internal Host	4,146 failed reverse DNS lookups (.in-addr.arpa) triggering NXDOMAIN bursts.	True Positive (Internal Network Reconnaissance)
192.168.202.140	Internal Workstation	Query length anomaly (>50 chars) targeting	False Positive (Local ISP Suffix Appending)

		Firefox update domains.	
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3. Chronology & Investigation Findings

3.1. Reconnaissance Activity (T1590.002)

- **Observation:** Host 192.168.202.83 generated repeated lookup failures against 44.206.168.192.in-addr.arpa.
- **Technical Analysis:** Excessive NXDOMAIN responses on reverse PTR lookups strongly indicate an automated discovery scan (such as nmap list scanning or automated port sweeps) attempting to map internal IP addresses to active hostnames.

3.2. Tunneling & Exfiltration Triage (T1048.003)

- **Observation:** Host 192.168.202.140 produced 16 queries with a length of 51 characters targeting versioncheck.addons.mozilla.org.hsd1.md.comcast.net.
- **Technical Analysis:** The query was flagged by length-based exfiltration filters. Investigation proved that a benign Firefox extension update check had the Comcast ISP search domain appended following an initial lookup failure. Classified as a **False Positive**.

3.3. Beaconing Evaluation (T1071.004)

- **Observation:** Host 10.10.117.210 accounted for over 18% of total network DNS traffic.
- **Technical Analysis:** Traffic consisted primarily of repetitive Microsoft Teredo IPv6-over-IPv4 queries. While operational, high beacon frequencies require strict monitoring to prevent uninspected covert tunneling.

4. Containment & Remediation Actions

- **Asset Containment:** Host 192.168.202.83 was flagged for endpoint isolation and triage to identify unauthorized scanning utilities or compromise.
- **Perimeter Hardening:** Enforced firewall egress filtering on UDP/TCP port 53, restricting outbound DNS communication exclusively to approved internal DNS forwarders.
- **Endpoint Hardening:** Recommended disabling transition protocols (Teredo/ISATAP) via Active Directory Group Policy (GPO) across endpoints not requiring IPv6 encapsulation.

5. Detection Engineering & Tuning

To prevent future alert fatigue and detect recurring activity:

1. Implemented correlation search SOC_ALERT_02_Reverse_DNS_Reconnaissance_Scan to trigger when failed PTR lookups exceed 1,000 events in 15 minutes.

2. Updated rule `SOC_ALERT_03_DNS_Tunneling_Exfiltration_Tuned` with domain exclusions (`*.comcast.net`, `*.local`) to eliminate recurring ISP suffix false positives.